# INDUSTRIAL METHANOL PRODUCTION PLANT

## Process Simulation and Material Balance Using DWSIM

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## Executive Summary

A methanol production process was modelled in DWSIM using the Peng-Robinson thermodynamic property package. The process uses a fresh synthesis-gas feed containing hydrogen and carbon monoxide, compression to elevated pressure, feed heating, an equilibrium reactor, product cooling, gas-liquid separation and recycle of unreacted gas. The recycle operation was reported by DWSIM as converged in two iterations.

The simulated liquid product stream has a reported methanol mole fraction of 0.99057338 (99.057 mol%). Based on the displayed product flow of 33.5448 kmol/h, the calculated methanol component flow is approximately 33.2286 kmol/h, corresponding to about 1,064.64 kg/h or 25.55 tonnes/day at continuous operation.

| Performance indicator | Value | Basis |
|---|---|---|
| Fresh syngas feed | 100 kmol/h | DWSIM feed stream |
| Reactor temperature | 251.85 °C | Isothermal reactor |
| Reactor pressure | 80 bar | Reactor operating pressure |
| Product temperature / pressure | 35 °C / 80 bar | Separator product stream |
| Methanol product purity | 99.057 mol% | Product mole fraction |
| Methanol production | 1,064.64 kg/h | Calculated from component molar flow |
| Recycle convergence | Yes; 2 iterations | Recycle block information |

*Note: The reported overall CO and $H_2$ conversions are approximate component-based calculations using the fresh feed and the displayed product stream. They assume no other external outlet streams (for example, purge, vent or drain). The material balance is therefore conditional on the stated process boundary.*

## 1. Introduction

Methanol is an important chemical intermediate and is widely used in the manufacture of formaldehyde, acetic acid, solvents, fuels and other chemicals. Industrial methanol synthesis commonly uses a hydrogen-rich synthesis gas and a catalytic conversion process. This project demonstrates the construction and solution of a simplified methanol synthesis flowsheet using process-simulation software.

## 2. Objectives

The objectives of this project are to develop a steady-state methanol process flowsheet, apply a suitable thermodynamic property package, simulate compression and heating of synthesis gas, represent methanol synthesis with an equilibrium reaction, cool and separate the reactor effluent, recycle the unreacted gas, and calculate key product and material-balance indicators.

## 3. Process Description

Fresh synthesis gas is supplied at 25 °C and 30 bar with a total molar flow of 100 kmol/h. Its nominal composition is 66.6667 mol% hydrogen and 33.3333 mol% carbon monoxide, giving an $H_2/CO$ ratio of approximately 2. The feed is compressed to 80 bar and mixed with the recycle gas. The combined feed is heated to 251.85 °C and sent to an isothermal equilibrium reactor.

The reactor effluent is cooled to 35 °C and sent to a gas-liquid separator operating at 80 bar. The liquid-rich outlet is taken as the methanol product stream, while the vapour outlet is returned to the feed mixer through a recycle block. The recycle block breaks the flowsheet calculation loop and iterates until the recycle stream is consistent with the process calculation.

### 3.1 Main Equipment

| Equipment / Stream | Function | Specified or Reported Condition |
|---|---|---|
| FRESH_SYNGAS | Fresh synthesis-gas feed | 100 kmol/h; 25 °C; 30 bar |
| C-101 | Feed compressor | Outlet pressure: 80 bar; efficiency: 75% |
| M-101 | Mixer | Combines compressed fresh feed and recycle return |
| E-101 | Feed heater | Outlet temperature: 251.85 °C |
| R-101 | Equilibrium reactor | Isothermal 251.85 °C, 80 bar; ΔP = 0 bar |
| E-102 | Product cooler | Outlet temperature: 35 °C |
| V-101 | Gas-liquid separator | 35 °C, 80 bar |

| Equipment / Stream | Function | Specified or Reported Condition |
|---|---|---|
| RCY-101 | Recycle logical operation | Converged: Yes; iterations taken: 2 |

## 4. Thermodynamic Model and Reaction

The Peng-Robinson (PR) equation of state was selected as the thermodynamic property package. The reaction set contains the gas-phase equilibrium reaction:

> $CO + 2\ H_2 \rightleftharpoons CH_3OH$

The equilibrium reactor was configured in isothermal mode at the inlet temperature. The reaction is represented as an equilibrium reaction rather than a kinetic reactor with an explicit catalyst-rate expression. Consequently, the reactor result represents the selected equilibrium model and assumptions, not a catalyst-specific industrial reactor prediction.

## 5. Input and Output Stream Data

The following values are transcribed from the displayed DWSIM stream results. Composition values are mole fractions. Minor differences can arise because the simulator screenshots show rounded values.

| Stream | P (bar) | T (°C) | Total flow (kmol/h) | Total mass flow (kg/h) |
|---|---|---|---|---|
| Fresh syngas | 30 | 25 | 100.0000 | 1,068.06 |
| Methanol product | 80 | 35 | 33.5448 | 1,068.03 |
| Recycle gas (internal) | 80 | 35 | 50.7532 | 415.078 |

| Component | Fresh feed mole fraction | Product mole fraction |
|---|---|---|
| Methanol | 0 | 0.99057338 |
| Water | 0 | 0 |
| Nitrogen | 0 | 0 |
| Carbon dioxide | 0 | 0 |
| Carbon monoxide | 0.333333 | 0.0030865468 |
| Hydrogen | 0.666667 | 0.0063400701 |

## 6. Material Balance and Calculations

The overall plant boundary is taken from the fresh syngas feed to the methanol product stream. The recycle gas is an internal stream and is not counted as an external feed or external product. This avoids double-counting the recycle flow.

### 6.1 Component Molar Flow

For each component *i*, the component molar flow is calculated from the total stream molar flow and its mole fraction:

> $F_i = F_{total} \times y_i$

Methanol component flow in the product:

> $F_{MeOH} = 33.5448 \times 0.99057338 = 33.2286$ kmol/h

### 6.2 Methanol Mass Production

Using the molecular weight of methanol, MW = 32.04 kg/kmol:

> $m_{MeOH} = 33.2286 \times 32.04 = 1{,}064.64$ kg/h

For continuous operation over 24 hours:
Daily production = 1,064.64 × (24 / 1,000) = **25.55 tonnes/day**

### 6.3 Overall CO Conversion

> CO in fresh feed = 100 × 0.3333333 = 33.3333 kmol/h
> CO in product = 33.5448 × 0.0030865468 = 0.10354 kmol/h
> $X_{CO}$ = [(33.3333 - 0.10354) / 33.3333] × 100 = **99.689%**

### 6.4 Overall $H_2$ Conversion

> $H_2$ in fresh feed = 100 × 0.666667 = 66.6667 kmol/h
> $H_2$ in product = 33.5448 × 0.0063400701 = 0.21268 kmol/h
> $X_{H2}$ = [(66.6667 - 0.21268) / 66.6667] × 100 = **99.681%**

*These overall conversion values assume the methanol product is the only external outlet containing unreacted CO and $H_2$. If a purge, vent or other outlet is present in the actual flowsheet, its component flow must be included in the overall balance.*

### 6.5 Recycle and Single-Pass Conversion

The displayed recycle gas flow is approximately 50.7532 kmol/h. The recycle-to-fresh-feed molar ratio is:
*R/F = 50.7532 / 100 = 0.508* kmol recycle per kmol fresh feed

Using the displayed recycle-gas CO mole fraction of 0.23065318, the recycle CO flow is approximately 11.7064 kmol/h. The reactor inlet CO flow is therefore approximately 33.3333 + 11.7064 = 45.0397 kmol/h. The reactor outlet CO flow, represented by CO in the product plus CO in the recycle gas, is approximately 0.10354 + 11.7064 = 11.8099 kmol/h.

> Single-pass CO conversion = [(45.0397 - 11.8099) / 45.0397] × 100 = **73.78%**

## 7. Overall Mass Balance Check

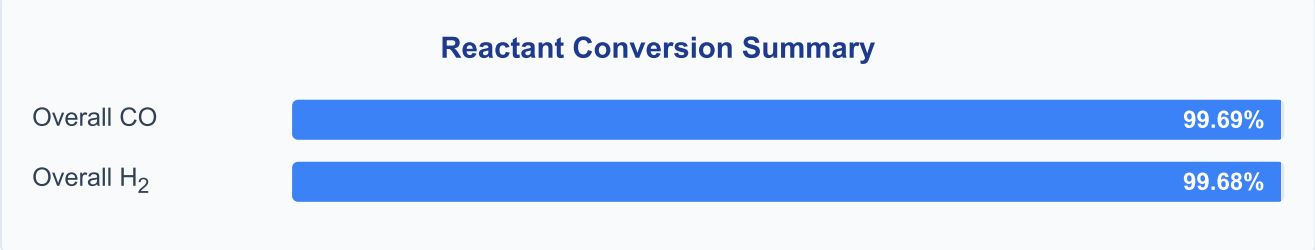
The displayed total mass flow rates are:

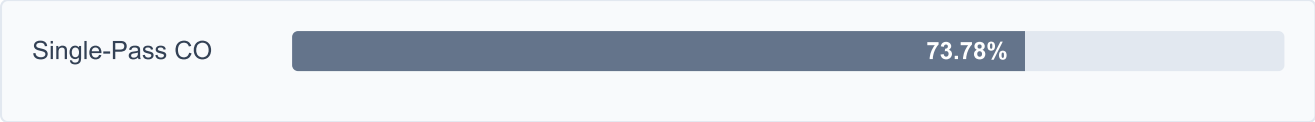
| Item | Mass flow (kg/h) |
|---|---|
| Fresh syngas feed | 1,068.06 |
| Methanol product | 1,068.03 |
| Difference (feed - product) | 0.03 |
| Difference relative to feed | 0.0028% |

The small difference between the displayed total mass flow rates is consistent with rounding of the displayed stream values. A rigorous component-wise material balance should be checked using full-precision DWSIM stream exports and by confirming that all external outlet streams are included.

## 8. Results and Discussion

The simulation produced a methanol-rich liquid product stream with a reported methanol mole fraction of 99.057%. The calculated methanol component production is approximately 1,064.64 kg/h. Cooling the reactor effluent to 35 °C enabled the separator to produce a liquid-rich methanol product and a vapour recycle stream. The recycle block reported successful convergence in two iterations, indicating that the recycle loop was solved for the displayed simulation case.






The calculated overall CO and $H_2$ conversions are high because the overall plant boundary includes the recycle of unreacted gas and assumes no separate purge or vent stream. These overall conversion values must not be confused with the single-pass reactor conversion, estimated at approximately 73.78% using the displayed recycle data.

## 9. Assumptions and Limitations

1. Steady-state operation is assumed.
2. Peng-Robinson is used as the thermodynamic property package.
3. The reactor is an equilibrium reactor, not a detailed kinetic/catalyst model.
4. The separator is represented as a simplified gas-liquid separator.
5. No explicit purge, vent, drain or off-spec stream is included in the reported external balance; if present, it must be added.
6. The calculations use rounded values transcribed from screenshots; full-precision stream exports are preferred for final validation.
7. Heat duties, compressor power, equipment sizing, catalyst design, heat integration, safety analysis and economic evaluation are outside the current scope.

## 10. Conclusion

A steady-state methanol synthesis flowsheet was developed and solved in DWSIM using the Peng-Robinson property package. The process includes feed compression, mixing, heating, equilibrium reaction, cooling, gas-liquid separation and recycle. The reported recycle block converged successfully in two iterations. The final product stream contains approximately 99.057 mol% methanol, with a calculated methanol production rate of about 1,064.64 kg/h (25.55 tonnes/day). The results demonstrate the use of process simulation for flowsheet development and preliminary material-balance evaluation.

## 11. Suggested Further Work

Future improvements may include using a kinetic reactor model with catalyst data, evaluating operating pressure and temperature effects, optimising recycle ratio and purge, performing heat-exchanger and compressor energy analysis, sizing major equipment, and conducting an economic assessment of methanol production.

## Appendix A: Key Equations

| Quantity | Equation |
|---|---|
| Component molar flow | $F_i = F_{total} \times y_i$ |
| Component mass flow | $m_i = F_i \times MW_i$ |
| Methanol production | $m_{MeOH} = F_{MeOH} \times MW_{MeOH}$ |
| Overall component conversion | $X = [(F_{i,fresh} - F_{i,product}) / F_{i,fresh}] \times 100$ |
| Recycle ratio | $R/F = F_{recycle} / F_{fresh}$ |
| Mass balance residual | $\Sigma m_{in} - \Sigma m_{out}$ |